National Testing Agency

Question Paper Name :	B Tech 8th Apr 2026 Shift 2
Subject Name :	B. Tech
Creation Date :	2026-04-08 18:49:45
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6911218
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69112143
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112143
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 691121526 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the relation R on the set $\{-2,-1, 0, 1, 2\}$ defined by $(a, b) \in R$ if and only if $1+ab > 0$. Then, among the statements :

I. The number of elements in R is 17

II. R is an equivalence relation

Options :

6911211786. Only I is true

6911211787. Only II is true

6911211788. Both I and II are true

6911211789. Neither I nor II is true

Question Number : 2 Question Id : 691121527 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of values of $z \in \mathbb{C}$, satisfying the equations

$|z-(4+8i)|=\sqrt{10}$ and $|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$, is :

Options :

6911211790. 0

6911211791. 2

6911211792. 1

6911211793. 4

Question Number : 3 Question Id : 691121528 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the system of linear equations :

$x+y+z=6,$

$x+2y+5z=10,$

$2x+3y+\lambda z=\mu$

has infinitely many solutions, then the value of $\lambda+\mu$ equals :

Options :

6911211794. 12

6911211795. 16

6911211796. 22

6911211797. 28

Question Number : 4 Question Id : 691121529 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\begin{bmatrix}\alpha & 1 & 2\\ 2 & 3 & 0\\ 0 & 4 & 5\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0 & 0\\ 0 & -5\alpha & 0\\ 0 & 4\alpha & -2\alpha\end{bmatrix}$ + adj(A). If $\det(B)=66$, then $\det(\text{adj}(A))$ equals :

Options :

6911211798. 289

6911211799. 361

6911211800. 441

6911211801. 529

Question Number : 5 Question Id : 691121530 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\alpha = 3+4+8+9+13+14+\ldots$ upto 40 terms. If $(\tan\beta)^{\frac{\alpha}{1020}}$ is a root of the equation $x^2+x-2=0$, $\beta \in \left(0, \frac{\pi}{2}\right)$, then $\sin^2\beta + 3\cos^2\beta$ is equal to :

Options :

6911211802. 2

6911211803. $\frac{7}{4}$

6911211804. $\frac{5}{2}$

6911211805. $\frac{3}{2}$

Question Number : 6 Question Id : 691121531 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the entire distance out of bus, scooter and car. The probabilities of the candidate going by bus, scooter and car, respectively, are $\frac{2}{5}$, $\frac{1}{5}$ and $\frac{2}{5}$. The probabilities that the candidate reaches late at the examination centre are $\frac{1}{5}$, $\frac{1}{3}$ and $\frac{1}{4}$ if the candidate uses bus, scooter and car, respectively. Given that the candidate reached late at the examination centre, the probability that the candidate travelled by bus is :

Options :

6911211806. $\frac{11}{37}$

6911211807. $\frac{12}{37}$

6911211808. $\frac{13}{37}$

6911211809. $\frac{14}{37}$

Question Number : 7 Question Id : 691121532 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1. Then, the variance of all these 10 observations is equal to :

Options :

6911211810. 5.96

6911211811. 6.14

6911211812. 6.04

6911211813. 6.24

Question Number : 8 Question Id : 691121533 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $26\left(\frac{2^3}{3}\left({}^{12}C_2\right)+\frac{2^5}{5}\left({}^{12}C_4\right)+\frac{2^7}{7}\left({}^{12}C_6\right)+\ldots+\frac{2^{13}}{13}\left({}^{12}C_{12}\right)\right)=3^{13}-\alpha$, then α is equal to :

Options :

6911211814. 45

6911211815. 48

6911211816. 51

6911211817. 54

Question Number : 9 Question Id : 691121534 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag is empty, is :

Options :

6911211818. 18

6911211819. 36

6911211820. 39

6911211821. 72

Question Number : 10 Question Id : 691121535 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If a straight line drawn through the point of intersection of the lines $4x+3y-1=0$ and $3x+4y-1=0$, meets the co-ordinate axes at the points P and Q, then the locus of the mid point of PQ is :

Options :

6911211822. $x+y-7=0$

6911211823. $x+y-14xy=0$

6911211824. $2x+y+14xy=0$

6911211825. $x+2y-14xy=0$

Question Number : 11 Question Id : 691121536 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let O be the vertex of the parabola $y^2=4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C, then the length of its latus rectum is :

Options :

6911211826. 1

6911211827. 2

6911211828. 4

6911211829. 8

Question Number : 12 Question Id : 691121537 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where inverse trigonometric functions take only the principal values.

Given below are two statements :

Statement I : $\cos(\alpha+\beta) > 0$.

Statement II : $\cos(\alpha) < 0$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211830. Both **Statement I** and **Statement II** are true

6911211831. Both **Statement I** and **Statement II** are false

6911211832. **Statement I** is true but **Statement II** is false

6911211833. **Statement I** is false but **Statement II** is true

Question Number : 13 Question Id : 691121538 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the function $f(x) = e^{\sin|x|} - |x|$, $x \in \mathbf{R}$, consider the following statements :

Statement I : f is differentiable for all $x \in \mathbf{R}$.

Statement II : f is increasing in $\left(-\pi, -\frac{\pi}{2}\right)$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211834. Both **Statement I** and **Statement II** are true

6911211835. Both **Statement I** and **Statement II** are false

6911211836. **Statement I** is true but **Statement II** is false

6911211837. **Statement I** is false but **Statement II** is true

Question Number : 14 Question Id : 691121539 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2\left(\vec{a} \times \vec{b}\right) + 3\left(\vec{b} \times \vec{c}\right) = \vec{0}$.

If $\vec{a} \cdot \vec{c} = 15$, then $\vec{c} \cdot \left(\hat{i} + \hat{j} - 3\hat{k}\right)$ is equal to :

Options :

6911211838. -6

6911211839. -5

6911211840. -4

6911211841. -3

Question Number : 15 Question Id : 691121540 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $\frac{x-4}{1} = \frac{y-9}{2} = \frac{z-5}{1}$ be the point $(1, \mu, 2)$. Then the distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$ and $\frac{x-\lambda}{2} = \frac{y-\mu}{3} = \frac{z+5}{6}$ is equal to :

Options :

6911211842. $\frac{12}{7}$

6911211843. $\frac{\sqrt{145}}{7}$

6911211844. $\frac{\sqrt{146}}{7}$

6911211845. $\frac{\sqrt{143}}{7}$

Question Number : 16 Question Id : 691121541 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}}{(\sqrt{x+1})(\sqrt{x^4+x^2+1})}\,dx$ is equal to :

Options :

6911211846. $\frac{1}{3}\log_e(3-2\sqrt{2})$

6911211847. $\frac{2}{3}\log_e(4+\sqrt{2})$

6911211848. $\frac{2}{3}\log_e(3+2\sqrt{2})$

6911211849. $\frac{1}{3}\log_e(1+6\sqrt{2})$

Question Number : 17 Question Id : 691121542 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $y=y(x)$ be the solution of the differential equation

$x\sqrt{1-x^2}\,dy + \left(y\sqrt{1-x^2}-x\cos^{-1}x\right)dx=0, x\in(0,1), \lim_{x\to 1^-} y(x)=1$. Then $y\left(\frac{1}{2}\right)$ equals :

Options :

6911211850. $3-\frac{\pi}{\sqrt{3}}$

6911211851. $4-\sqrt{3}\,\pi$

6911211852. $4 - \frac{2\pi}{\sqrt{3}}$

6911211853. $3 - \frac{\pi}{2\sqrt{3}}$

Question Number : 18 Question Id : 691121543 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f:(1,\infty)\to\mathbf{R}$ be a function defined as $f(x)=\frac{x-1}{x+1}$. Let $f^{i+1}(x)=f\left(f^{i}(x)\right)$, $i=1, 2,...,25$, where $f^1(x)=f(x)$. If $g(x)+f^{26}(x)=0$, $x\in(1,\infty)$, then the area of the region bounded by the curves $y=g(x)$, $2y=2x-3$, $y=0$ and $x=4$ is :

Options :

6911211854. $\frac{1}{8}+\log_e 2$

6911211855. $\frac{1}{4}+\log_e 2$

6911211856. $\frac{5}{6}+3\log_e 2$

6911211857. $\frac{5}{6}+\log_e 2$

Question Number : 19 Question Id : 691121544 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)=\begin{cases}\frac{1}{3} & , \ x\le \pi/2\\ \frac{b(1-\sin x)}{(\pi-2x)^2} & , \ x>\pi/2\end{cases}$. If f is continuous at $x=\pi/2$, then the value of $\int_0^{3b-6}|x^2+2x-3|\,dx$ is :

Options :

6911211858. 5

6911211859. 2

6911211860. 3

6911211861. 4

Question Number : 20 Question Id : 691121545 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\frac{x^2}{f(a^2+7a+3)}+\frac{y^2}{f(3a+15)}=1$ represent an ellipse with major axis along y-axis, where f is a strictly decreasing positive function on $\mathbf{R}$. If the set of all possible values of a is $\mathbf{R}-[\alpha,\beta]$, then $\alpha^2+\beta^2$ is equal to :

Options :

6911211862. 28

6911211863. 40

6911211864. 61

6911211865. 24

Mathematics Section B

Section Id :	69112144
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112144
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 691121546 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The sum of squares of all the real solutions of the equation $\log_{(x+1)}(2x^2+5x+3)=4-\log_{(2x+3)}(x^2+2x+1)$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 691121547 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\int_{\pi/6}^{\pi/4}\left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx=\alpha\log_e(\sqrt{3}-1)$, then $9\alpha^2$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 691121548 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let a line L_1 pass through the origin and be perpendicular to the lines

$L_2 : \vec{r} = (3 + t)\hat{i} + (2t - 1)\hat{j} + (2t + 4)\hat{k}$ and

$L_3 : \vec{r} = (3 + 2s)\hat{i} + (3 + 2s)\hat{j} + (2 + s)\hat{k}$, $t, s \in \mathbf{R}$.

If (a, b, c), $a \in \mathbf{Z}$, is the point on L_3 at a distance of $\sqrt{17}$ from the point of intersection of L_1 and L_2, then $(a+b+c)^2$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 691121549 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the circle $C : x^2 + y^2 - 6x - 8y - 11 = 0$. Let a variable chord AB of the circle C subtend a right angle at the origin. If the locus of the foot of the perpendicular drawn from the origin on the chord AB is the circle $x^2 + y^2 - \alpha x - \beta y - \gamma = 0$, then $\alpha + \beta + 2\gamma$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 691121550 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let f be a polynomial function such that

$\log_2(f(x)) = \left(\log_2\left(2 + \frac{2}{3} + \frac{2}{9} + \ldots\ldots \infty\right)\right) \cdot \log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$, $x > 0$ and $f(6) = 37$. Then $\sum_{n=1}^{10} f(n)$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	69112145
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory